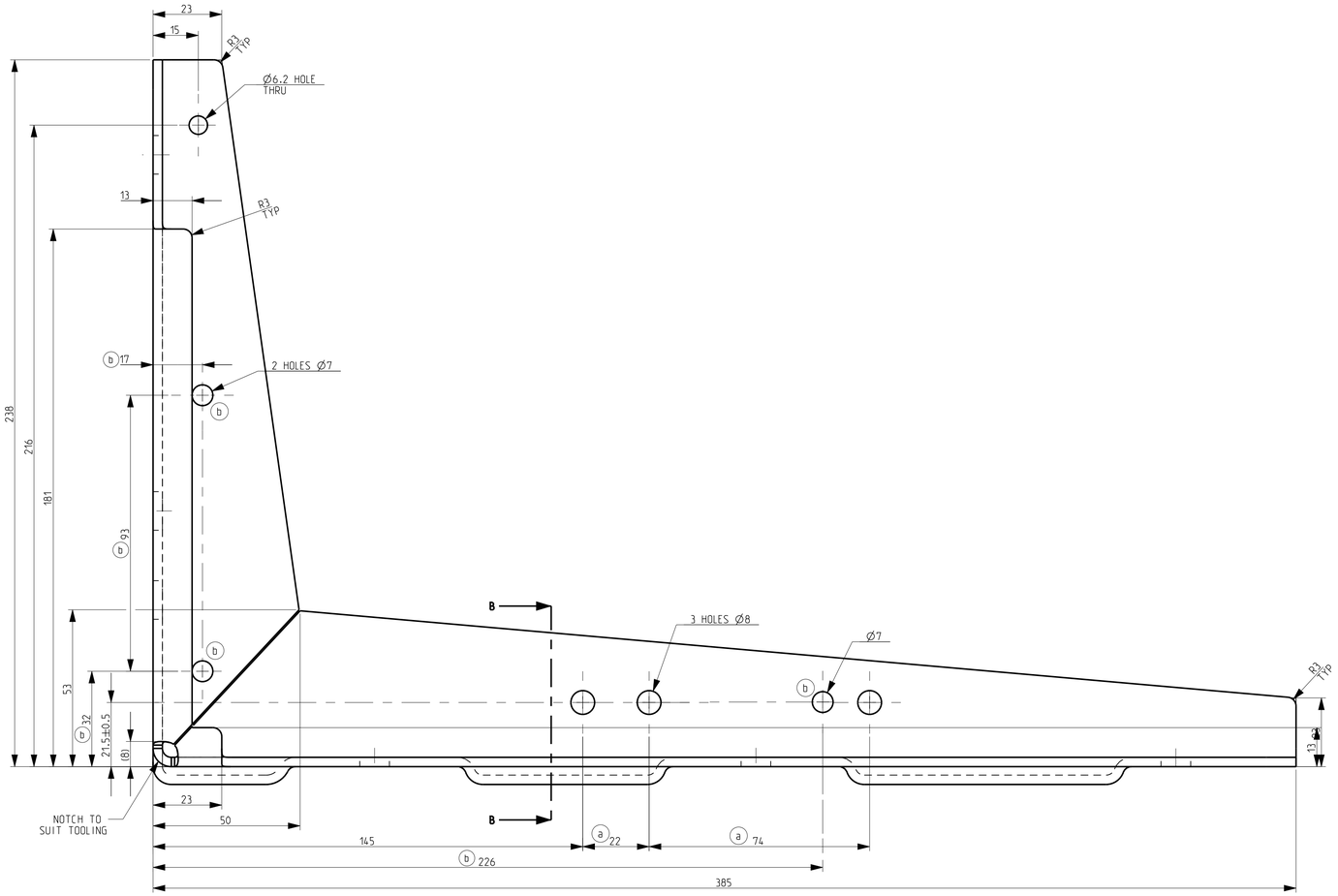
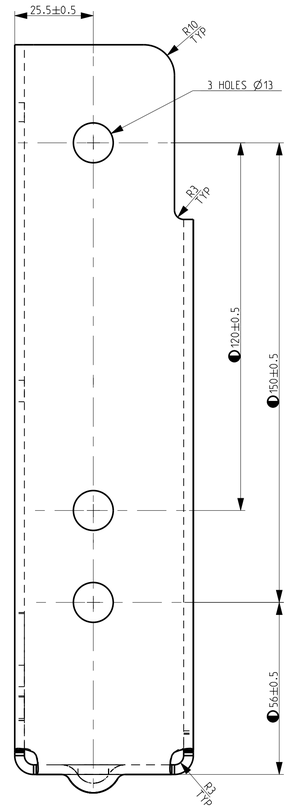
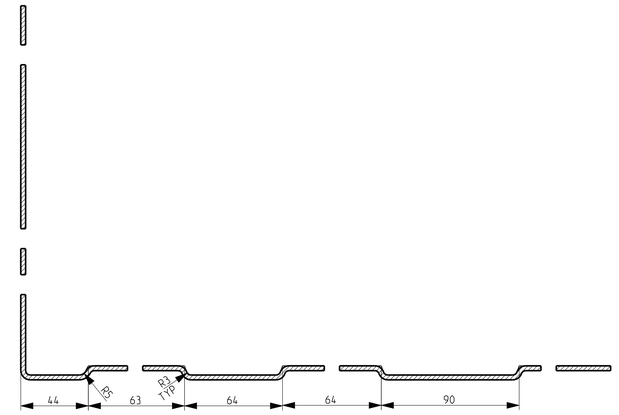
DR4

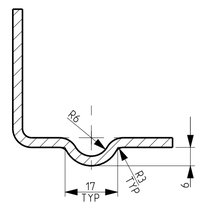
| Do not scale the drawing. | Unless otherwise stated. 1. All dimensions in mm. | Nominal Dimension | Tolerance as per IS 2102 (Part 1):1993 |
|---|---|---|---|
| If in doubt, ask. | 2. Drawing drawn to first angle projection | - | - |



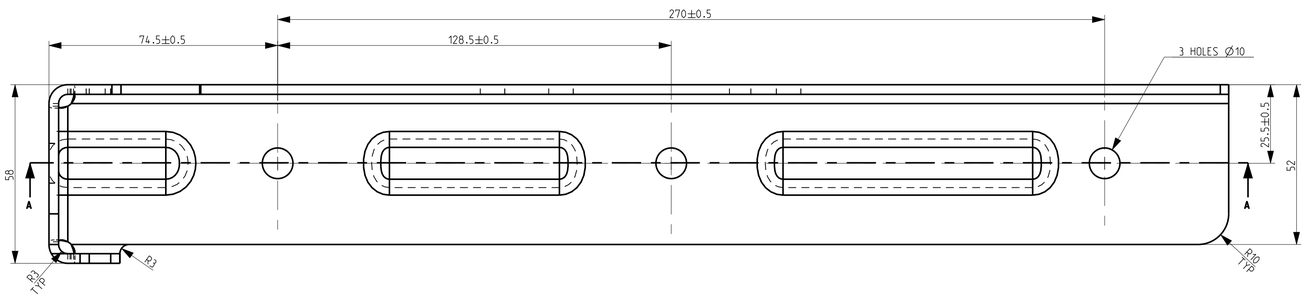




SECTION A-A
SCALE 1:2




SECTION B-B




N12
DEBURR

| Tolerances of stampings mfg. by forming as per TS 12130 Standard | | | | | | | |
|---|---|---|---|---|---|---|---|
| Nominal Size Range | Accuracy Grade | From 0.1 to 1 | Over 1 Up to 3 | Over 3 Up to 6 | Over 6 Up to 10 | Over 10 |
| From 1 To 6 | f | ±0.2 | ±0.3 | ±0.6 | —— | —— |
| | m | ±0.4 | ±0.6 | —— | —— | —— |
| | g | ±0.6 | —— | —— | —— | —— |
| | sg | —— | —— | —— | —— | —— |
| Over 6 Up to 10 | f | ±0.4 | ±0.6 | ±0.8 | ±1 | —— |
| | m | ±0.6 | ±1 | ±1.2 | ±1.5 | —— |
| | g | ±1.2 | ±2 | ±2.5 | —— | —— |
| | sg | ±1.6 | ±2 | ±2.5 | —— | —— |
| Over 10 Up to 25 | f | ±0.6 | ±0.8 | ±1 | ±1.2 | ±1.5 |
| | m | ±1 | ±1.2 | ±1.5 | ±2 | ±2.5 |
| | g | ±2 | ±2.5 | ±4 | —— | —— |
| | sg | ±2.5 | ±4 | ±4 | —— | —— |
| Over 25 Up to 63 | f | ±0.8 | ±1 | ±1.2 | ±1.5 | ±2 |
| | m | ±1.2 | ±1.5 | ±2 | ±2.5 | ±4 |
| | g | ±2.5 | ±4 | ±6 | —— | —— |
| | sg | ±4 | ±6 | ±6 | —— | —— |
| Over 63 Up to 160 | f | ±1 | ±1.2 | ±1.5 | ±2 | ±2.5 |
| | m | ±1.5 | ±2 | ±2.5 | ±4 | ±6 |
| | g | ±4 | ±6 | ±10 | ±15 | ±25 |
| | sg | ±6 | ±10 | ±10 | ±15 | ±25 |
| Over 160 Up to 400 | f | ±1.2 | ±1.5 | ±2 | ±2.5 | ±3 |
| | m | ±2 | ±2.5 | ±4 | ±6 | ±10 |
| | g | ±6 | ±10 | ±15 | ±25 | ±40 |
| | sg | ±10 | ±15 | ±15 | ±25 | ±40 |
| Over 400 Up to 1000 | f | ±1.6 | ±2 | ±2.5 | ±3 | ±4 |
| | m | ±2.5 | ±4 | ±6 | ±10 | ±15 |
| | g | ±10 | ±15 | ±25 | ±40 | ±60 |
| | sg | ±15 | ±25 | ±25 | ±40 | ±60 |

| SURFACE ROUGHNESS SYMBOLS / VALUES IN µm Ra | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Symbol | N12 | N11 | N10 | N9 | N8 | N7 | N6 | N5 to N1 |
| Value | 50 | 25 | 6.3 | 1.6 | 0.8 | 0.1 | 0.8 | 0.8 |

FOR OTHER ROUGHNESS SYMBOLS & VALUES. SEE IS 10719

| General Tolerance on machining as per ISO 2768-1mmedium class of TS12115 | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Dimension | Over | 0.5 | 6 | 30 | 120 | 400 | 1000 | 2000 |
| | Upto & incl | 6 | 30 | 120 | 400 | 1000 | 2000 | 4000 |
| Tolerance | | ±0.1 | ±0.2 | ±0.3 | ±0.5 | ±0.8 | ±1.2 | ±2 |

For all Vehicle Components & Assemblies.
Tata Group Mark & Part No. to be marked as per TS 1080s

For M1 & N1 class of vehicles, applicable standards are: TS1114.,TS1118.,TS1119 & TS1120

*Material description,size,spec,std No.*

3.15mm Thk - T SH HR-Fe410 TS23521

*Classification of characteristics:*
- ● *Critical.* ▲ *Key instruction*
- ◐ *Major.* ◯ *(None) Minor*
- ● *Applies to upper limit*
- ● *Applies to lower limit*

*This drawing is the sole property of Tata Motors Limited. It should not be copied or communicated to any person without the written approval of Tata Motors Limited.*

TATA MOTORS LIMITED
ERC - PUNE

For explanation of drawing format contents, refer to TS11103
8845 x130 7067 PCF - 072(92) 6/2011

| Date | Mod. No. | Sl No. | Zone | Modification | Chkd | Appd |
|---|---|---|---|---|---|---|
| 01/07 2019 | 18PM M57 001 | b | D4.E3 D1.C1 | 1 HOLE Ø7 ADDED WITH DIM 226<br>2 HOLES Ø7 ADDED WITH DIM. 32. 93 AND 17 | PDT | RMP |
| 06/06 2018 | 12 PP 254 163 | a | E3 E4 | DIMNS 42 AND 85 WERE 22 AND 74 | SRP | RMP |
| 01/10 2015 | 12 PP 242 010 | NR | - | NEW RELEASE | NMM | HMP |

| | 2015 | SIGN | DATE |
|---|---|---|---|
| Drawn | SDB | | 11/09 |
| Chkd | NMM | | 11/09 |
| Appd | HMP | | 12/09 |

Scale: 1:1 (1:2)

Envelope Dimensions: 386.28 x 244.09 x 58.30
Final mass in kg.: 1.318

Product/Group No.: ULTRA 1518

Drg/Part Designation: BRACKET,BATTERY MTG,RH

| | |
|---|---|
| Surface protection as per standard | SEE UPPER LEVEL ASSY |
| Tolerance as per standard | - |
| Replaces Drg | - |
| Reference Drg No. | - |
| Opposite Hand Drg/Part No. | 5530 3154 33 01N |
| Drg/Part No. | 5530 3154 33 02 |
| Sheet No.....1..... of .....1..... Sheets | |

Original drawing size A1